

Prenatal PFAS Exposure in a Twin Pregnancy Cohort: Associations with Thyroid Disruption, Placental Malperfusion, and Fetal Growth Restriction

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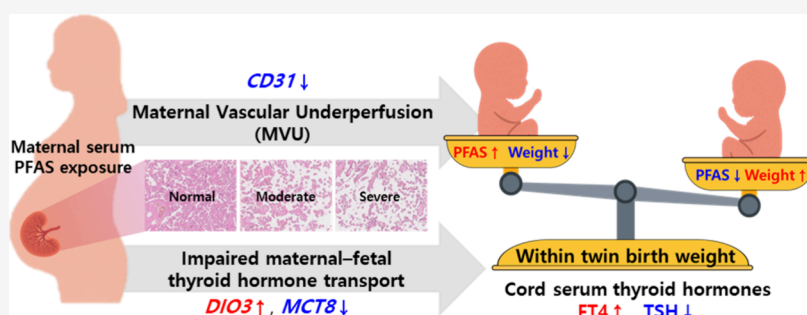
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ABSTRACT: Twin pregnancies are particularly vulnerable to chemical exposures such as per- and polyfluoroalkyl substances (PFAS), yet mechanistic evidence linking PFAS to adverse pregnancy outcomes remains limited. In this study, we examined maternal and cord serum PFAS exposure in relation to thyroid function, placental pathology, and fetal growth in a subset of twins from the Ideal Breastmilk (IBM) cohort in Korea (78 mothers, 199 neonates). Cord serum PFAS exposure was associated with a thyroid hormone profile characterized by higher FT4 and lower TSH within the reference range. In addition, higher cord PFOA, PFNA, and Σ PFCA concentrations were associated with lower birth weight. Maternal concentrations of these PFAS were highest in pregnancies with severe placental malperfusion (maternal vascular underperfusion, MVU). Placental gene expression analysis revealed reduced *CD31* and *MCT8* expression and elevated *DIO3* expression in association with PFAS exposure and pathology, suggesting impaired placental vascular-endocrine function and disrupted maternal–fetal thyroid hormone transport. Together, these findings provide mechanistic evidence linking PFAS exposure to fetal growth impairment in twin pregnancies. Our observations highlight the heightened vulnerability of twin pregnancies to environmental chemical exposures and underscore the need for strengthened exposure mitigation strategies during pregnancy.

KEYWORDS: PFAS, twin pregnancy, placenta, thyroid hormones, birth weight

1. INTRODUCTION

Per- and polyfluoroalkyl substances (PFASs) have been widely used in industrial and consumer products due to their durability and resistance to water, oil, and stains. However, these substances are highly persistent in the environment and the human body, leading to their ubiquitous detection.^{1,2} PFAS exposure is associated with various endocrine and metabolic disturbances, including thyroid hormone disorders and alterations in cholesterol levels, in the general population.^{3,4} For instance, exposure to the PFAS mixture was positively related to the odds of thyroid hormone disorder, as well as to increased levels of both high-density lipoprotein (HDL) cholesterol and total cholesterol, among U.S. adults.⁴

PFAS elimination in humans varies with compound-specific physicochemical properties, particularly carbon chain length and functional group (e.g., carboxylate vs sulfonate).⁵ In

women, elimination occurs through several pathways, including urine, feces, and, to a lesser extent, menstrual blood loss.⁶ Because menstruation ceases during pregnancy, the absence of this elimination pathway may reduce overall clearance, potentially contributing to increased maternal body burden and enhanced transplacental transfer to the fetus.^{7,8} This mechanism may be particularly relevant in twin pregnancies due to increased physiological demands and maternal hemodynamic alterations.⁹ In particular, the greater expansion

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of maternal blood volume and potentially elevated glomerular filtration rate (GFR) in twin pregnancies may alter the volume of distribution and renal clearance of PFAS, thereby potentially influencing maternal–fetal body burden dynamics.¹⁰ During pregnancy, PFAS exposure is known to disrupt endocrine regulation, particularly thyroid function, and may interact with placental hormones that are critical for maintaining endocrine and metabolic homeostasis.^{11,12} Prenatal PFAS exposure has been consistently associated with reduced birth weight in multiple epidemiological studies.^{13,14} Reduced birth weight is a key marker of impaired fetal development and has been linked to adverse health outcomes later in life.^{15–17}

In recent years, the incidence of twin pregnancies has increased primarily due to the widespread use of assisted reproductive technologies (ARTs), particularly among women of advanced maternal age.¹⁸ Twin pregnancies undergo distinct physiological changes compared to singleton pregnancies, substantially influencing the hormonal and metabolic environment of both the mother and the twin fetuses.¹⁹ Additionally, their unique physiological and hormonal milieu may alter the toxicokinetic behavior of environmental chemicals relative to singleton pregnancies.²⁰

Emerging evidence suggests that PFAS may disrupt placental vascularization and gene expression, potentially impairing maternal–fetal communication and contributing to fetal endocrine dysregulation. However, data on the associations among PFAS exposure, placental function, and fetal thyroid or growth outcomes remain limited, particularly in twin pregnancies, which are inherently at a higher risk for placental insufficiency.

To address this gap, we investigated the associations of maternal and cord serum PFAS exposure with key pregnancy outcomes and placental pathology, focusing on twin pregnancies. The twin design enables within-pair comparisons to account for shared genetic and intrauterine factors, thereby improving inference regarding the potential causal effects of maternal environmental exposures.²¹ In addition, we explored placental molecular mediators, including thyroid-related gene expression, to provide insights into potential biological pathways linking PFAS exposure to fetal thyroid function and growth. Through this approach, we hypothesized that PFAS exposure would be associated with altered fetal thyroid function and impaired placental function, leading to reduced fetal growth in twin pregnancies.

2. MATERIALS AND METHODS

2.1. Study Design and Participants

A subset of twin pregnant women ($n = 78$) from the Ideal Breastmilk (IBM) cohort ($n = 442$) and their neonatal twins ($n = 199$, some unmatched with mothers due to insufficient maternal samples) were selected based on the availability of biological specimens for chemical analysis and data analysis (Figure 1). The IBM cohort is a prospective birth cohort recruited from Seoul National University Hospital between June 2016 and December 2019. Its inclusion and exclusion criteria, follow-up protocol, and data collection have been previously described.^{22,23} The study was conducted in accordance with the Helsinki Declaration and was approved by the Institutional Review Board of Seoul National University Hospital (IRB no. 1512–039–727 for mothers, and 1606–117–772 for children). All participating mothers provided written informed consent.

2.2. Measurement of PFAS and Related Compounds

A total of 33 PFAS, including precursors, alternatives, and isomers, were analyzed in maternal (average 27.0 ± 5.9 weeks of gestation)

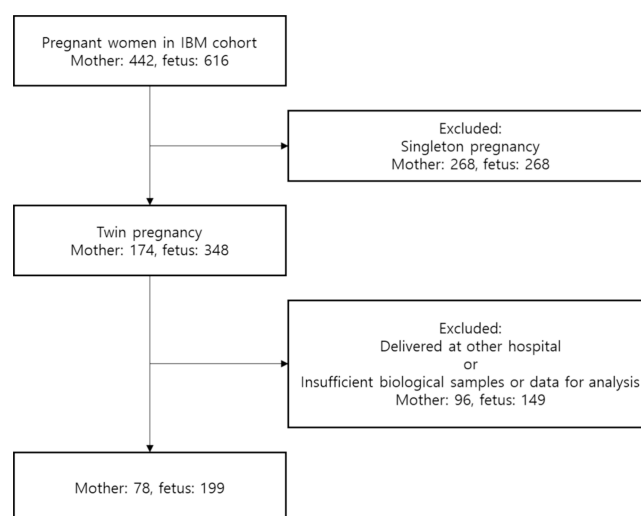


Figure 1. Selection of twin pregnancies and neonates from the Ideal Breastmilk (IBM) cohort.

and cord serum (at birth) samples using ultrahigh-performance liquid chromatography–tandem mass spectrometry (UHPLC–MS/MS; 1290 Infinity II, Agilent, USA; API 4000, SCIEX, USA) following solid-phase extraction.^{8,24} Calibration and quantification were conducted for 30 target analytes using 16 isotope-labeled internal standards (Tables S1 and S2). Detailed sample pretreatment and instrumental parameters are provided in Supporting Information (Tables S3 and S4). All analytes exhibited excellent linearity ($r^2 > 0.995$), with recoveries and precision (RSD < 20%) within acceptable ranges (Table S5). Quantification was based on signal ratios of each analyte to its corresponding internal standard. Limits of detection (LODs) were determined as 3.3 times the standard deviation of the intercept divided by the slope of the calibration curve, and nondetectable results are reported in Table S5.

2.3. Thyroid Hormones Analysis

Cord serum was measured for free thyroxine (FT4) and free triiodothyronine (FT3) by radioimmunoassay (RIA) using the RIAKEY Free T4 RIA Tube (Shin Jin Medics Inc., Goyang, Republic of Korea) and the BRAHMS FT3 RIA Kit (Thermo Scientific, Hennigsdorf, Germany), respectively. Thyroid-stimulating hormone (TSH) levels in serum were quantified by using an immunoradiometric assay (Shin Jin Medics Inc.). All assays were performed using an automatic γ counter (Wizard² 2470, PerkinElmer, USA). The intra- and interassay coefficients of variation were <10%, confirming analytical reliability.

Because specific reference intervals for twin neonates have not been established, interpretation of cord-blood thyroid hormone values was based on reference ranges for term singleton cord blood reported in previous studies: TSH (0.4–12.0 mIU/L), FT4 (1.2–2.6 ng/dL), and FT3 (0.12–0.26 ng/dL).^{25,26}

2.4. Placental Pathology and Transcriptional Changes

Placenta sampling was performed following the 2016 Amsterdam Placental Workshop Group Consensus Statement.^{27,28} At least three full-thickness parenchymal sections were obtained from the central two-thirds and paracentral regions of the placental disc, encompassing the architecture from the chorionic plate to the basal plate,²⁹ along with one additional section containing the umbilical cord and chorioamniotic membrane roll. All hematoxylin–eosin–stained slides from these sections (≥ 4 sections per placenta) were systematically reviewed, and maternal vascular underperfusion (MVU) features were assessed across all examined sections rather than a single representative field. All slides were evaluated by a board-certified perinatal pathologist blinded to clinical outcomes, except for gestational age, which was essential for accurate interpretation.

Each placenta was systematically examined for four MVU features: (1) villous infarction, (2) accelerated villous maturation with increased syncytial knots, (3) increased intervillous fibrinoid deposition, and (4) distal villous hypoplasia. A semiquantitative scoring system assigned one point for each MVU lesion present, generating a maternal underperfusion score (range: 0–4). Placentas were classified as follows: score 0 = normal; score 1 = mild MVU; and scores ≥ 2 = severe MVU.

Twin pairs enrolled in the IBM cohort were categorized by placental histopathologic findings: Both Normal ($n = 27$ pairs): both placentas without malperfusion; Discordant ($n = 36$ pairs): one placenta with malperfusion and the cotwin placenta normal; Both malperfusion ($n = 15$ pairs): both placentas demonstrating malperfusion features.^{30,31}

2.5. Placental Transcriptional Changes

To quantify placental transcriptional changes, a stratified subset of pregnancies was constructed from the 78 twin pregnancies with available placental tissues. Pregnancies with severe MVU were first identified among samples with frozen tissue that yielded RNA of acceptable quality after extraction and quality control. To minimize confounding by developmental maturity and fetal growth, candidate pregnancies in the mild MVU and normal groups were selected so that the distributions of gestational age at birth and birth weight were comparable to those observed in the severe MVU group. Within these candidate pools, pregnancies were randomly selected to represent each pathological category while avoiding overrepresentation of extreme values. This resulted in a balanced subset consisting of normal placentation ($n = 7$), mild MVU ($n = 6$), and severe MVU ($n = 6$), with comparable gestational age across groups (normal: 36.3 ± 0.8 weeks; mild: 36.7 ± 0.5 weeks; severe: 35.5 ± 0.8 weeks, $p > 0.05$) and birth weight (normal: 2.6 ± 0.5 kg; mild: 2.6 ± 0.4 kg; severe: 2.2 ± 0.7 kg, $p > 0.05$).

For these stratified subsets of pregnancies, placental samples collected from the maternal side of the cotyledon (after snap-freezing in liquid nitrogen and storage at -80°C) were used for RNA extraction. Total RNA was isolated using TRIzol reagent (Invitrogen, USA) following the manufacturer's protocol, and RNA concentration and purity were verified spectrophotometrically (A260/A280 ratio > 1.8). Complementary DNA was synthesized from $1\ \mu\text{g}$ of total RNA using the ReverTra Ace qPCR RT Master Mix (Toyobo, Japan). Quantitative real-time PCR (RT-qPCR) was performed using Power SYBR Green PCR Master Mix (Applied Biosystems, USA) on a QuantStudio 5 Real-Time PCR System (Thermo Fisher Scientific, USA). *CD31* (*PECAM1*), *DIO3*, *MCT8* (*SLC16A2*), and β -actin were quantified (Table S6). The relative expression was normalized to *GAPDH* as the housekeeping gene and calculated using the $2^{-\Delta\Delta C_t}$ method.³² All reactions were performed in duplicate.

2.6. Statistical Analysis

Demographic characteristics between the full IBM twin cohort and the twin subcohort included in this study were compared using nonparametric tests for continuous variables and chi-square tests for categorical variables.

For PFAS detected in $\geq 70\%$ of the samples and for the single nondetected TSH measurement, values below the LOD were imputed as LOD divided by $\sqrt{2}$.³³ PFAS concentrations were converted to molar concentrations ($\mu\text{mol/L}$).

Analyses of infant outcomes were conducted at two levels: individual-level and within-pair analyses. For individual-level analyses, generalized estimating equation (GEE) models were used to examine associations between PFAS exposure and outcomes, accounting for within-pair correlation by specifying the twin pair as the clustering unit. For within-pair analyses, differences in exposure and outcome between cotwins were calculated and analyzed using generalized linear models (GLMs). Because each twin pair contributed a single difference value, the observations were treated as independent.

PFAS concentrations and thyroid hormone levels were natural log-transformed prior to analysis to reduce skewness and approximate normal distribution. All models were consistently adjusted for maternal age, education level, parity, prepregnancy body mass index

(BMI), gestational age at sampling, gestational age at birth, and fetal sex. Although maternal PFAS concentrations were unavailable for some mother–neonate pairs due to insufficient sample volume ($n = 22$, see Figure 1 and Table S7), other maternal characteristics, including maternal age, education level, parity, and prepregnancy BMI, were complete for all participating mothers and were included as covariates in the adjusted models. For analyses of thyroid hormone outcomes, urinary iodine concentration (UIC), normalized to urinary creatinine, was further included as a covariate. In sensitivity analyses, we also adjusted for placental type (or chorionicity) and pregnancy-related complications, including gestational diabetes and gestational hypertension. Placental type was determined by prenatal ultrasonography during the first or second trimester based on the number of placental masses and the presence of the 'lambda' (twin-peak) sign or T-sign.^{34,35}

Associations between maternal PFAS concentrations and MVU severity scores (0–2, from normal to severe) were evaluated using the Kruskal–Wallis test. In discordant pairs, both the mean and the higher severity scores within each pair were considered. Spearman's correlation analyses were used to assess relationships between PFAS concentrations and outcome measures. Because maternal PFAS concentrations were shared within each twin pair, data were duplicated for each individual placenta, and GEE models were applied to account for within-pair correlation.

All regression analyses were conducted using a single-chemical, single-outcome framework, and p -values < 0.05 were considered statistically significant. To account for multiple testing across all PFAS exposures and outcomes, false discovery rate (FDR) adjustment was applied using the Benjamini–Hochberg method. Adjusted p -values were reported as q -values, with $q < 0.05$ considered statistically significant.

Associations between PFAS exposure and placental gene expression (*CD31*, *DIO3*, and *MCT8*) were evaluated using Spearman's correlation analysis. To explore independent contributions, linear regression models including both PFAS levels and MVU severity (categorized as a binary variable: mild vs severe) were constructed. Given the small sample size and potential multicollinearity between PFAS exposure and MVU severity, these analyses were primarily intended to be exploratory. Multicollinearity was assessed using variance inflation factors (VIFs) to ensure the stability of the regression estimates. All statistical analyses were performed in R (version 4.0.2).

3. RESULTS

3.1. Maternal and Twin Infant Characteristics and PFAS Concentrations

Most participating mothers were 36 years old or older and were not obese (Table 1).

The mean birth lengths and weights of the infants were 46.6 cm and 2.47 kg, respectively, with a slightly higher proportion of male infants observed. No significant differences in demographic characteristics were detected between the twin pairs used in this study and the IBM twin cohort (Table S7). Details of the demographic characteristics of the mothers ($n = 78$) are available elsewhere.⁸

In the cord serum samples, PFOA, PFNA, PFDA, and L-PFOS exhibited the highest detection frequencies and median concentrations (Table 2). PFOA was the most prevalent perfluoroalkyl carboxylic acid (PFCA), found in 99.5% of fetuses, with a median concentration of $1.77\ \mu\text{g/L}$. Other commonly detected substances included PFNA (99.5%) and PFUnDA (96.98%). Among perfluorosulfonic acids (PFSAs), L-PFOS had the highest detection frequency (99.5%), with a median concentration of $1.13\ \mu\text{g/L}$. The most frequently detected PFAS precursors were 6:2 diPAP (52.76%) and 8:2 diPAP (51.76%). In contrast, several PFCAs, such as PFPeA,

Table 1. Demographic and Clinical Characteristics of Mothers and Twin Infants in the Study Population

		<i>n</i> (%) or mean $\pm$ SD ^a
mothers (<i>n</i> = 78)		
age (years)	<30	4 (5.1)
	30–36	34 (43.6)
	≥ 36	40 (51.3)
Prepregnancy BMI (kg/m ²)	<18.5	8 (10.3)
	18.5–23	40 (51.3)
	23–25	11 (14.1)
	>25	19 (24.4)
parity	nulliparous	60 (76.9)
assisted reproductive technology	yes	68 (88.5)
placental type	monochorionic diamniotic (MCDA)	5 (6.4)
	dichorionic diamniotic (DCDA)	78 (93.6)
smoking status during pregnancy	no	78 (100)
alcohol consumption during pregnancy	no	78 (100)
education	$\leq$ high school	7 (9.0)
	college/university	56 (71.8)
	>university	15 (19.2)
UIC ^b (μ g/g creatinine)		493.01 $\pm$ 1101.7
infant twins (<i>n</i> = 199)		
sex	male	108 (54.3)
sex composition of twin pairs	MM ^c	29 (29.3)
	FF ^d	20 (20.2)
	MF ^e	50 (50.5)
gestational age at birth	weeks	36.3 $\pm$ 1.0
preterm birth (<37 weeks)	N (%)	104 (52.3)
birth size	height (cm)	46.6 $\pm$ 2.1
	weight (kg)	2.5 $\pm$ 0.4
	head circumference (cm)	33.2 $\pm$ 1.4

^aSD, standard deviation. ^bUIC, urinary iodine concentration normalized to urinary creatinine. ^cMM, male–male twin pairs. ^dFF, female–female twin pairs. ^eMF, sex-discordant pairs.

PFHxA, and PFDoDA, and PFSA, including PFBS, L-PFNS, and PFDS, showed low detection frequencies (<10%).

Most cord serum samples had thyroid hormone levels within normal reference ranges. Among the 199 infants included in the study, 38 (19%) exhibited TSH concentrations exceeding the upper reference limit of 12 μ IU/mL.

3.2. Associations of Cord Serum PFAS Exposure

3.2.1. Cord Serum Thyroid Hormones. Cord serum PFAS exposure was associated with a hormonal profile characterized by higher FT4 and lower TSH levels in cord serum (Figure 2). These associations remained statistically significant after additional adjustment for placental type and pregnancy-related complications. When analyses were stratified by preterm birth status, the associations were more pronounced and statistically significant in the preterm group (<37 weeks of gestation) (Table S8). Moreover, most associations remained significant even following the FDR adjustment ($q < 0.05$; Table S9) and were not materially altered when creatinine-normalized UIC was included as an additional covariate (Figure S1). Most associations also remained significant when analyses were restricted to infants within the reference TSH range ($\leq 12 \mu$ IU/mL) (Table S10).

For TSH, significant negative correlations were found for the summed PFCA ($\beta = -0.17$, 95% CI: -0.29 to -0.06 , $p < 0.01$) and summed PFSA ($\beta = -0.17$, 95% CI: -0.30 to -0.04 , $p = 0.01$). Among individual PFAS, statistical significance was observed only for PFOA ($\beta = -0.15$, 95% CI: -0.25 to -0.05 , $p < 0.01$), with most compounds showing negative but nonsignificant associations.

For FT4, significant positive correlations were found for most compounds, except for Br-PFHxS and L-PFHxS. The summed PFCA ($\beta = 0.04$, 95% CI: 0.00 to 0.07 , $p = 0.03$) and the total sum of all PFAS ($\beta = 0.04$, 95% CI: 0.00 to 0.08 , $p = 0.03$) were all positively associated with FT4. In tertile-based analyses assessing linear trends, FT4 also showed consistent, statistically significant linear associations.

For FT3, negative associations were generally observed with cord serum PFAS concentrations, although congener-specific heterogeneity was evident. Both Br-PFOS ($\beta = -0.09$, 95% CI: -0.16 to -0.01 , $p = 0.03$) and L-PFOS ($\beta = -0.09$, 95% CI: -0.16 to -0.01 , $p = 0.03$) were significantly associated with lower FT3 levels. In contrast, Br-PFHxS showed a positive trend, with higher FT3 levels in T2 and T3 than in T1 (reference) (Table S11).

3.2.2. Birth Weight. Cord serum PFAS exposure showed significant associations with decreased birth weight (Figure 3a). A negative direction of associations was found for PFOA ($\beta = -0.02$, 95% CI: -0.03 to -0.004 , $p = 0.02$), PFNA ($\beta = -0.11$, 95% CI: -0.21 to -0.01 , $p = 0.03$), and Σ PFCA ($\beta = -0.02$, 95% CI: -0.03 to -0.003 , $p = 0.02$).

Within twin pairs, larger differences in cord PFAS concentrations were associated with greater differences in birth weight (Figure 3b). Although statistical significance was lost after FDR adjustment in individual-level analyses, it was retained for most PFAS (except Σ PFAS) in within-pair analyses (Table S9). Significant positive β estimates were observed for PFOA ($\beta = 0.06$, 95% CI: 0.03 to 0.10 , $p < 0.01$), PFNA ($\beta = 0.36$, 95% CI: 0.14 to 0.57 , $p < 0.01$), and Σ PFCA ($\beta = 0.06$, 95% CI: 0.03 to 0.09 , $p < 0.01$), and the direction of association remained consistent for most other PFAS. This pattern was further supported by within-pair regression models (Figure 3c–e), which showed that greater differences in PFAS concentrations between cotwins were associated with greater decreases in body weight.

3.3. Associations of Maternal PFAS Exposure

3.3.1. Cord Serum Thyroid Hormones and Birth Weight. Maternal PFAS exposure was associated with lower TSH and higher FT4 levels, consistent with the association observed for cord serum PFAS concentrations (Figure 4). Similar to the findings for PFAS in cord serum, a more pronounced inverse association with TSH was observed in the preterm group (Table S12). Additional sensitivity analyses adjusting for placental type and pregnancy complications, including gestational diabetes mellitus and gestational hypertension, resulted in minimal changes in the observed associations, supporting the robustness of these findings. Overall, maternal exposure to the summed PFAS showed significant negative associations with TSH ($\beta = -0.24$, 95% CI: -0.43 to -0.04) and positive associations with FT3 ($\beta = 0.13$, 95% CI: 0.01 to 0.26). These associations remained directionally consistent, and effect estimates were not substantially altered when analyses were additionally adjusted for creatinine-normalized UIC (Figure S2).

Table 2. Concentrations of PFAS, Their Precursors, and Alternatives, and Thyroid Hormones in Umbilical Cord Serum of Infant Twins ($n = 199$; $\mu\text{g/L}$ unless Otherwise Noted)

PFAS	LOD ^a	DF ^b (%)	med (IQR ^c)	min-max
perfluoroalkyl carboxylic acids (PFCAs)				
PFBA	0.087	29.2	ND ^d (ND-0.17)	ND-2.69
PFPeA	0.074	3.5		ND-0.27
PFHxA	0.187	1.0		ND-0.27
PFHpA	0.047	9.1		ND-0.12
PFOA	0.049	99.5	1.77 (1.21–2.53)	ND-7.78
PFNA	0.035	99.5	0.39 (0.29–0.50)	ND-1.31
PFDA	0.032	94.0	0.09 (0.06–0.13)	ND-0.34
PFUnDA	0.061	97.0	0.21 (0.14–0.28)	ND-0.70
PFDoDA	0.089	23.1		ND-0.21
PFTTrDA	0.163	50.8	0.16 (ND-0.23)	ND-1.02
PFTeDA	0.082	11.1		ND-0.17
perfluorosulfonic acids (PFSAs)				
PFBS	0.087	20.1		ND-1.19
Br-PFHxS	0.064	86.9	0.31 (0.17–0.43)	ND-1.30
L-PFHxS	0.12	97.0	1.72 (0.92–2.54)	ND-4.42
Br-PFOS	0.107	95.5	0.26 (0.18–0.40)	ND-0.96
L-PFOS	0.065	99.5	1.13 (0.76–1.71)	ND-5.83
L-PFNS	0.066	1.0		ND-0.10
PFDS	0.048	0.5		ND-0.13
PFPeS	0.237	0.0		
L-PFHpS	0.507	0.0		
precursors				
FOSA	0.064	0.5		ND-0.14
N-MeFOSAA	0.078	2.5		ND-0.30
N-EtFOSAA	0.092	0.0		
4:2FTS	0.077	0.0		
6:2FTS	0.325	2.5		ND-78.90
8:2FTS	0.159	0.0		
6:2diPAP	0.184	52.8	0.20 (ND-0.38)	ND-1.44
6:2;8:2diPAP	0.099	1.0		ND-0.18
8:2diPAP	0.049	51.8	0.05 (ND-0.19)	ND-0.91
PFAS alternatives				
Gen-X	0.081	1.0		ND-1.99
ADONA	0.068	0.0		
9Cl-PF3ONS	0.107	28.6	ND (ND-0.12)	ND-0.39
11Cl-PF3OUdS	0.086	0.0		
thyroid hormones ^e				
FT3 (ng/dL)		100.0	0.21 (0.18–0.24)	0.06–0.36
FT4 (ng/dL)		10.0	1.20 (1.11–1.29)	0.83–1.57
TSH ($\mu\text{IU/mL}$)		99.5	7.31 (5.97–10.26)	ND-42.44

^aLOD, limit of detection. ^bDF, detection frequency. ^cIQR, interquartile range. ^dND, not detected. ^eSample sizes for FT3, FT4, and TSH were 193, 193, and 191, respectively.

Substance-specific analyses revealed similar directions of associations. For TSH, most of PFCAs, PFOA ($\beta = -0.17$, 95% CI: -0.29 to -0.05 , $p < 0.01$), PFNA ($\beta = -0.18$, 95% CI: -0.36 to -0.01 , $p = 0.04$), PFUnDA ($\beta = -0.21$, 95% CI: -0.41 to -0.01 , $p = 0.04$), summed PFCA ($\beta = -0.24$, 95% CI: -0.43 to -0.06 , $p = 0.01$), and 9Cl-PF3ONS ($\beta = -0.29$, 95% CI: -0.53 to -0.05 , $p = 0.02$) showed significant negative associations. For FT4, most substances exhibited positive correlations, with statistically significant associations for 6:2 diPAP ($\beta = 0.04$, 95% CI: 0.00 to 0.08, $p = 0.03$), and the strongest association with FT4 was observed for 9Cl-PF3ONS, the major compound of F-53B ($\beta = 0.07$, 95% CI: 0.02 to 0.12, $p = 0.01$). For FT3, most PFAS displayed positive correlations. Significant positive associations were observed for PFUnDA ($\beta = 0.11$, 95% CI: 0.01 to 0.21, $p = 0.04$), summed PFCA ($\beta =$

0.12, 95% CI: 0.01 to 0.22, $p = 0.03$), L-PFHxS ($\beta = 0.14$, 95% CI: 0.04 to 0.24, $p < 0.01$), and summed PFSA ($\beta = 0.13$, 95% CI: 0.00 to 0.26, $p = 0.05$), with summed PFAS showing a notably strong positive association ($\beta = 0.13$, 95% CI: 0.01 to 0.26, $p = 0.04$).

Maternal PFAS exposure was not significantly associated with individual fetal weight (Figure 4). However, most PFCAs, such as PFNA, PFDA, and PFUnDA, showed inverse but nonsignificant associations with birth weight.

3.3.2. Maternal Vascular Underperfusion (MVU). Representative histopathological findings corresponding to each MVU severity category are shown in Figure 5. Higher maternal PFAS exposure was associated with greater severity of MVU (Figure 6). When the higher severity score within each twin pair was applied (pair-level analyses), maternal PFAS

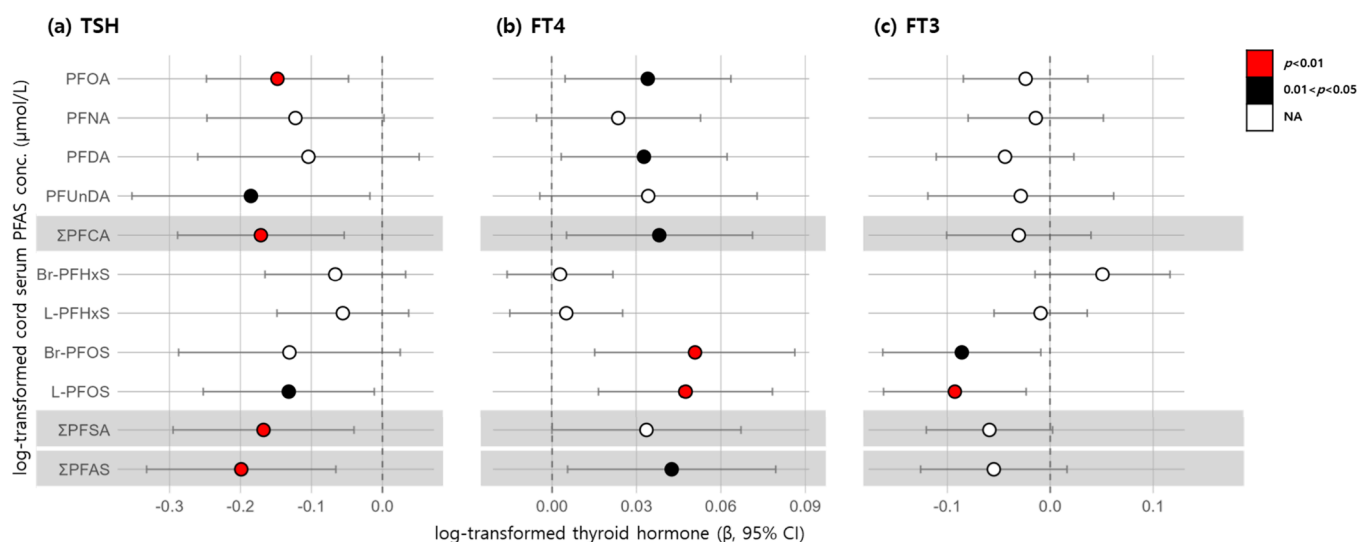


Figure 2. Association of cord serum PFAS concentrations with cord serum thyroid hormones: (a) TSH, (b) FT4, (c) FT3. Associations were analyzed using GEE models to account for clustering within twin pairs and were adjusted for maternal age, education, parity, prepregnancy BMI, gestational age, birth week, and fetal sex. PFAS concentrations and thyroid hormone levels were natural log-transformed prior to analysis. Red symbols indicate $p \leq 0.01$; black symbols indicate $0.01 < p \leq 0.05$; NA: not significant. Σ PFCA, Σ PFSA, and Σ PFAS represent the total molar concentrations of PFCAs, PFASs, and all PFAS detected in $\geq 70\%$ of the samples.

concentrations showed a positive association with the MVU severity score, indicating a dose–response relationship between maternal PFAS burden and the extent of placental vascular pathology (Figure 6a–c). When the analysis was limited to concordant pairs (i.e., twins with the same severity score), maternal PFOA and PFNA showed statistically significant odds ratios (ORs) for severe MVU (Table S13, OR ≈ 3.1 and 2.0, respectively).

Among the 19 placental samples included in the gene expression analysis, five corresponding infants had fetal TSH levels $>12 \mu\text{IU/mL}$. Of these, four were classified as having MVU (mild, $n = 3$; severe, $n = 1$), and one had no evidence of MVU.

RT-PCR analysis of placental gene expression stratified by MVU severity demonstrated significant correlations between transcript levels and severity (Figure 6d–f). Spearman's correlation analysis further indicated that *CD31* and *MCT8* expression levels were negatively correlated with MVU severity, whereas *DIO3* expression was positively correlated (Figure S3).

In addition to stratification by MVU severity, when we examined direct associations between maternal serum PFOA levels and placental gene expression, Spearman's correlation analysis revealed that PFOA concentrations were inversely correlated with *CD31* and *MCT8* expression, while the association with *DIO3* did not reach statistical significance (Table S14). Consistent with the primary analysis, MVU severity was strongly associated with increased *DIO3* expression and decreased *MCT8* expression. In mutually adjusted linear regression models, *CD31* tended to be associated with PFOA levels, whereas *DIO3* expression was primarily associated with MVU severity. For *MCT8*, both PFOA exposure and MVU severity showed significant correlations; however, only MVU retained borderline significance after mutual adjustment, suggesting a more prominent role of underlying placental pathology in this subset.

Multicollinearity was observed between PFOA and MVU severity (variance inflation factor ≈ 5); the direction and

magnitude of the effect size estimates remained consistent across models, suggesting that the interpretation of the exploratory findings was not substantially biased.

4. DISCUSSION

4.1. PFAS Exposure and Thyroid Hormone in Twin Pregnancy

The relatively high levels of PFAS exposure observed among the mothers and twins warrant attention. During pregnancy, reduced excretion pathways among women may promote PFAS accumulation in the maternal body and facilitate transplacental transfer to the fetus.⁸ Similar patterns of maternal PFAS accumulation and transplacental transfer have also been reported in singleton pregnancies in China.^{6,36} Maternal exposure levels of PFOA, PFOS, and PFHxS observed in the present population were generally higher than those reported in comparable populations from other countries, and cord serum PFHxS levels exceeded previously documented values in the literature (Table 2).⁸

Both cord and maternal serum PFAS levels were associated with modest but statistically significant increases in FT4 and decreases in TSH levels in cord serum, reflecting a population-level shift toward higher FT4 and lower TSH concentrations (Figures 2 and 4). Although thyroid hormone alterations in relation to PFAS exposure have been reported in several studies, evidence from twin populations remains limited. Given that thyroid hormone reference intervals are primarily derived from term singleton populations and may not be directly applicable to twin or predominantly preterm infants, these findings should be interpreted as relative changes within the physiological range rather than as evidence of clinically overt thyroid dysfunction or hormone excess.

The only available twin data are from the Chinese mother–infant Wuhan Twin Birth Cohort (WTBC), sampled between 2016 and 2018.^{31,37} Among the Chinese twins, in contrast to our observation, positive associations of cord plasma concentrations of PFOA, PFDA, PFUnDA, and PFHxS with TSH were reported.³¹ These divergent findings may reflect the

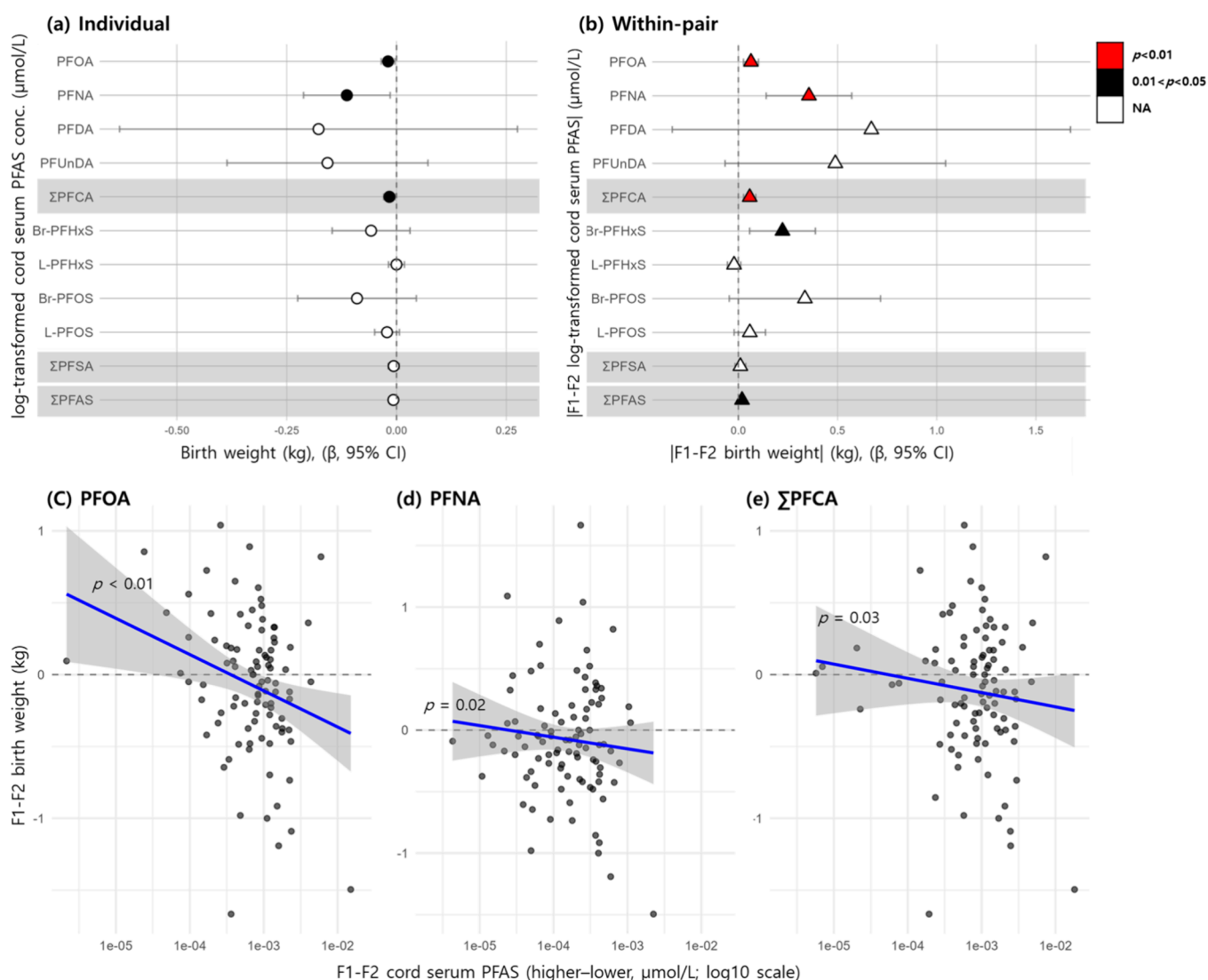


Figure 3. Associations of cord serum PFAS concentrations with birth weight. (a) Individual-level analysis using GEE models among all fetuses ($n = 199$) and (b) within-pair analysis using GLM based on absolute twin differences ($|F_1 - F_2|$, $n = 99$ pairs). Models were adjusted for maternal age, education level, parity, prepregnancy BMI, gestational age at sampling, gestational age at birth, and fetal sex. In panels (a) and (b), red symbols indicate $p \leq 0.01$; black symbols indicate $0.01 < p \leq 0.05$; NA, not significant. Σ PFCA, Σ PFSA, and Σ PFAS represent the total molar concentrations of PFCAs, PFSA, and all PFAS detected in $\geq 70\%$ of the samples, respectively. Panels (c–e) show within-pair differences in cord serum PFAS concentrations (higher–lower) and corresponding birth weight differences for (c) PFOA, (d) PFNA, and (e) Σ PFCA. The x-axis represents PFAS concentration differences (log scale), and the y-axis represents birth-weight differences (kg). Each point indicates one twin pair, and the blue lines indicate linear regression with shaded 95% confidence intervals.

difference in PFAS exposure levels (generally higher in this study) or maternal age (36 years in this study vs 30 years in WTBC) but warrant further investigations in other populations. However, in another Chinese cohort involving singleton pregnancies delivered by prelabor cesarean section, PFNA, PFDA, and PFOS showed negative associations with TSH, consistent with the current findings in Korean twins.³⁸

More pronounced inverse associations were observed in the preterm group for both maternal and cord serum PFAS concentrations (Tables S8 and S12). These stronger associations may reflect the incomplete maturation of the hypothalamic–pituitary–thyroid (HPT) axis in preterm infants. This developmental immaturity, coupled with dependence on maternal thyroid hormones during late gestation, may lower the threshold for PFAS-induced disruption and make their thyroid homeostasis more vulnerable to environmental

perturbations.^{39,40} Given the substantially higher rate of preterm birth in twins, the physiological immaturity of the HPT axis in these infants may increase their susceptibility to PFAS-related thyroid disruption.^{40,41} This higher prevalence of prematurity may, therefore, partly explain the more pronounced associations observed between PFAS exposure and thyroid hormone levels in this population.

4.2. PFAS Exposure and Growth Retardation in Twins

Elevated cord serum PFOA and PFNA concentrations were associated with lower birth weight (Figure 3), consistent with prior findings from singleton populations. Several previous studies, including those from China and Denmark, have reported impaired fetal growth associated with PFAS exposure, supporting our findings of significant inverse associations between cord serum PFOA, PFNA, and Σ PFCA concentrations and birth weight (Figure 3a,c–e).^{42–44}

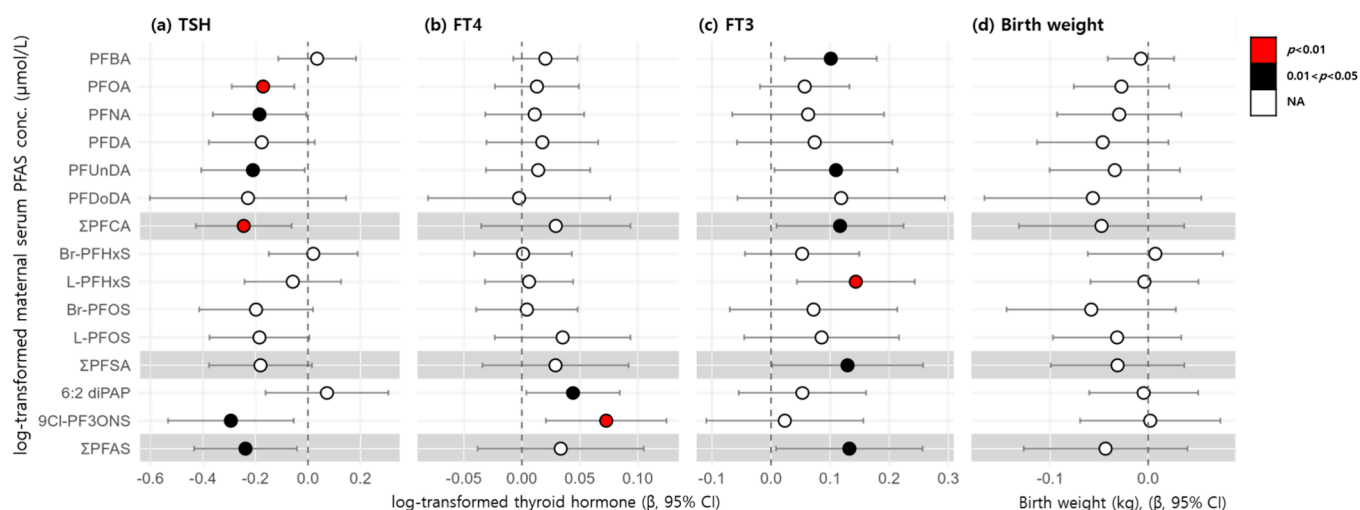


Figure 4. Association of maternal serum PFAS concentrations ($n = 78$) with fetal outcomes ($n = 156$). Associations were analyzed using GEE models adjusted for maternal age, education level, parity, prepregnancy BMI, gestational age at sampling, gestational age at birth, and fetal sex. (a) TSH, (b) FT4, (c) FT3, and (d) birth weight. PFAS concentrations and thyroid hormone levels were natural log-transformed prior to analysis. Red symbols indicate $p \leq 0.01$; black symbols indicate $0.01 < p \leq 0.05$; NA: not significant. Σ PFCA, Σ PFSA, and Σ PFAS represent the total molar concentrations of PFCAs, PFSAs, and all PFAS detected in $\geq 70\%$ of the samples, respectively.

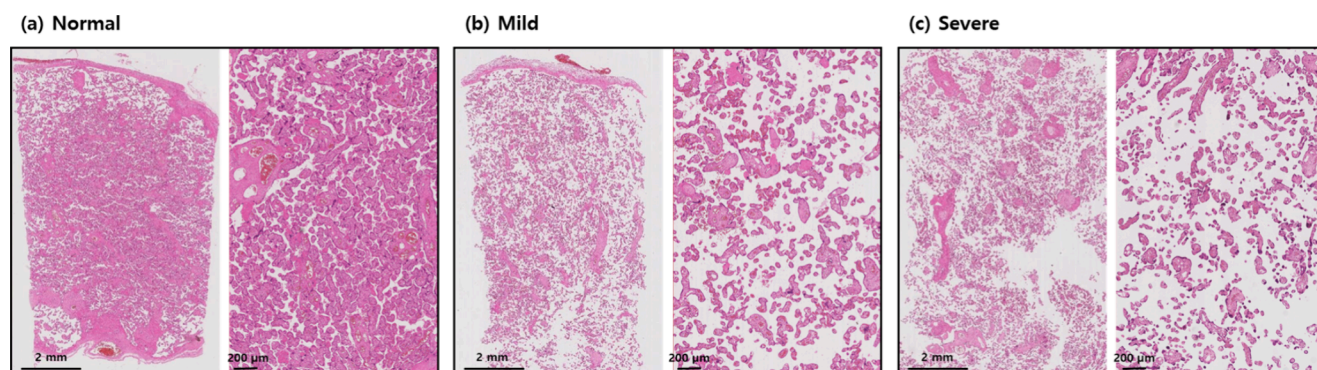


Figure 5. Representative placental histopathological features: (a) Normal placenta showing well-formed terminal villi with sparse syncytial knots. (b) Mild maternal vascular underperfusion (MVU), characterized by focal villous infarction and increased syncytial knotting. (c) Severe MVU, showing extensive villous infarction, accelerated villous maturation, and diffuse fibrinoid necrosis. MVU was scored (0–4) based on four features, i.e., villous infarction, accelerated villous maturation, fibrinoid deposition, and distal villous hypoplasia, and was classified as normal (0), mild (1), or severe (≥ 2). Scale bars: low magnification, 2 mm; high magnification, 200 μ m.

Similar findings were reported in the WTBC cohort of 204 twin neonates, where differences in PFAS exposure contributed to in utero birth-weight discordance.³⁷ By effectively controlling shared genetic and intrauterine factors, such within-pair analyses provide robust evidence for a potential causal link, which is further supported by biological plausibility.^{31,45} PFAS exposure during pregnancy may impair fetal growth through metabolic and vascular dysfunction. PFAS can activate peroxisome proliferator-activated receptors (PPARs) and alter liver fatty acid-binding protein (L-FABP) expression, potentially disrupting glucose metabolism, insulin sensitivity, and lipid homeostasis.⁴⁶ Epidemiological studies in singleton pregnancies have reported positive associations between maternal PFAS concentrations and the risk of GDM and gestational hypertension.⁴⁷ Epidemiological studies further suggest alterations in amino acid, lipid/fatty acid, bile acid, and androgen metabolism, as well as changes in placental arterial vasculature and resistance.^{48–51} PFAS may also interfere with thyroid hormone homeostasis, representing another potential pathway linking exposure to fetal growth restriction.⁵²

4.3. MVU Linking Thyroid Hormones and Growth Restriction

Significant associations between maternal PFOA, PFNA, and Σ PFCA levels and MVU severity (Figure 6) may in part explain the growth restriction observed among twins. PFAS have been reported to perturb the placental environment, including its vascular structure and function, thereby influencing fetal growth.⁵¹ Additionally, our findings suggest that PFAS exposure may disrupt placental endocrine regulation, including thyroid hormone homeostasis, potentially contributing to impaired fetal growth and development.⁵² Consistent with this possibility, findings from the Shanghai-Minhang Birth Cohort Study (S-MBCS) singleton pregnancy cohort showed associations between maternal PFAS concentrations and altered DNA methylation of placental genes involved in thyroid hormone transport and signaling, suggesting that disrupted placental TH regulation may be a potential mechanism.⁵³

The observed association between maternal but not cord PFAS concentrations and MVU severity likely reflects differences in exposure timing and pathophysiological

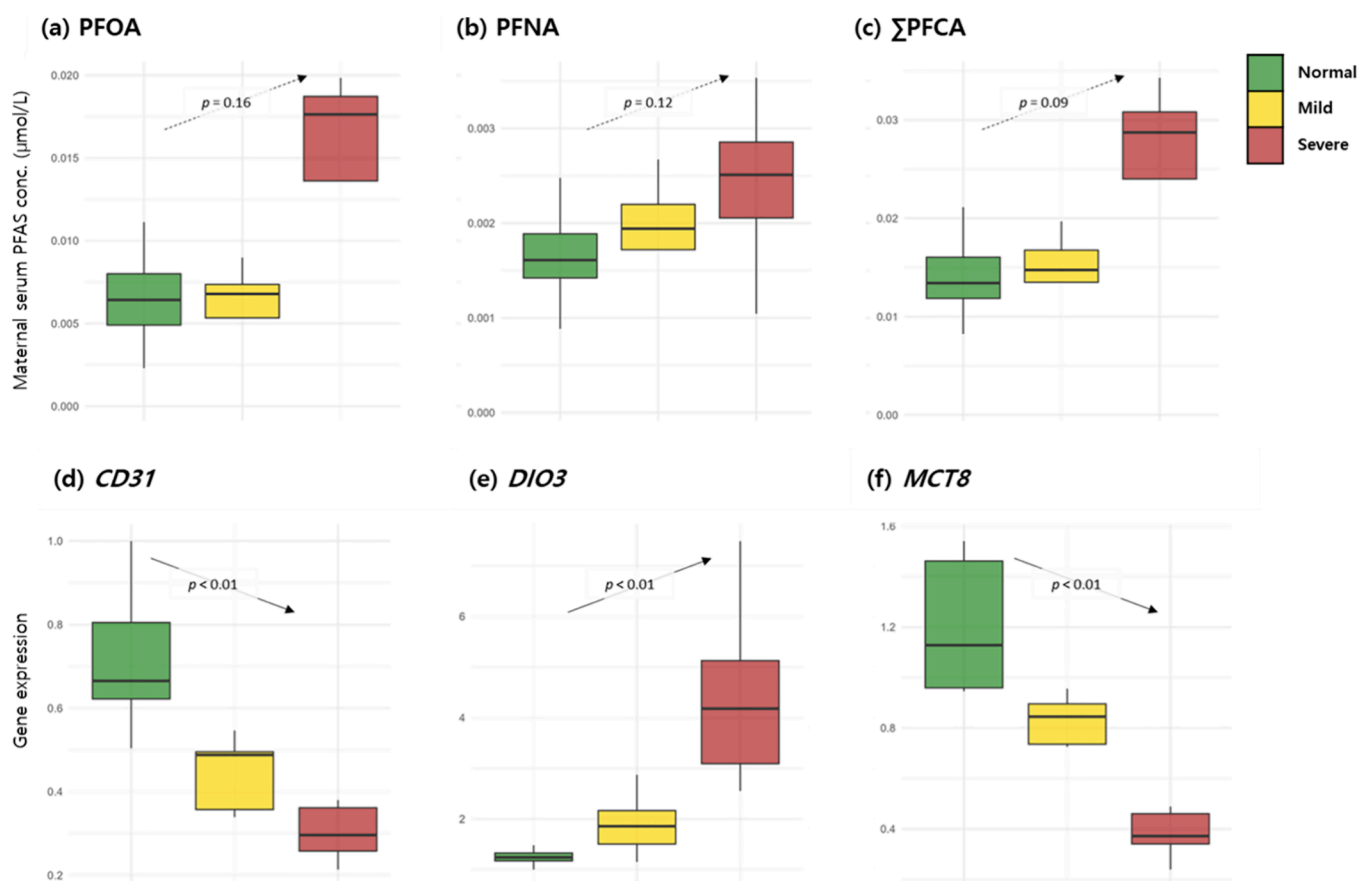


Figure 6. Associations of maternal serum concentrations with MVU severity and placental gene expression. Panels (a–c) show maternal serum concentrations of (a) PFOA, (b) PFNA, and (c) Σ PFCA according to pair-level MVU severity, defined by the maximum severity observed within each twin pair: 0, both normal (green, $n = 27$); 1, mild lesion in either placenta (yellow, $n = 32$); and 2, severe lesion in at least one placenta (red, $n = 19$). Panels (d–f) compare gene expression levels of (d) *CD31*, (e) *DIO3*, and (f) *MCT8* across MVU severity groups in the transcriptomic subset: normal (Group 0, green, $n = 6$), mild (Group 1, yellow, $n = 7$), and severe (Group 2, red, $n = 6$). Boxes represent the interquartile range with the median; whiskers indicate the range. p -values were obtained using the Kruskal–Wallis test.

relevance. Maternal PFAS levels represent chronic exposure that can influence placental development from early gestation, which is a critical window for the establishment of the placental vascular architecture. In contrast, cord serum concentrations reflect a complex interplay of maternal exposure, placental transfer efficiency, and fetal pharmacokinetics, specifically at the time of delivery. As such, cord PFAS levels may not sufficiently capture the cumulative exposure necessary to induce placental vascular pathologies.

In this study, cord serum levels of PFOA, PFNA, and Σ PFCA showed stronger negative correlations with birth weight (Figure 3), and their maternal concentrations were higher in pregnancies with greater MVU severity (Figure 6). Notably, when both concordant and discordant pairs were analyzed together, these associations became more pronounced, suggesting that elevated PFAS levels may exacerbate placental pathology discordance in twin pregnancies.⁴⁸

The observed associations between MVU severity and PFAS exposure, such as PFOA, PFNA, and Σ PFCA, suggest that placental dysfunction may represent a potential pathway linking maternal PFAS exposure to impaired cord serum thyroid hormone regulation and impaired fetal growth (Figure 6). It should be noted that transcriptional changes in *CD31*, *MCT8*, and *DIO3* may not represent direct consequences of PFAS exposure and could instead reflect downstream responses of placental pathology. In this study, all three

genes showed associations with MVU severity, supporting their relevance to placental structural and functional alterations. However, because transcriptomic analyses were conducted in a small subset of placentas selected to represent each MVU category, PFAS exposure levels and pathological severity may be closely coupled, limiting the ability to disentangle their independent effects. While this sampling design enhances the contrast between pathological states, it may also introduce collinearity between PFAS exposure and placental pathology, warranting cautious interpretation.

Reduced expression of *CD31* and *MCT8*, together with increased expression of *DIO3*, in placentas with more severe malperfusion supports the hypothesis that PFAS exposure impairs placental function through interlinked vascular and endocrine pathways. *CD31* expression showed a more predominant association with PFAS levels, whereas *DIO3* expression was more closely associated with MVU severity. *MCT8* exhibited a mixed profile, suggesting contributions from both PFAS-associated and pathological processes. Functionally, downregulation of *MCT8* indicates reduced thyroid hormone transport to the fetus, whereas upregulation of *DIO3* suggests enhanced inactivation of thyroid hormones, further limiting the hormonal availability to the developing fetus. *CD31*, an endothelial marker involved in placental angiogenesis, was also reduced, indicating impaired vascularization.⁵⁴ Similarly, in S-MBCS, PFOA was associated with

decreased placental *SLCO1C1* methylation, suggesting impaired placental thyroid hormone transport.⁵³

Importantly, although 19% of infants had elevated cord TSH levels (>12 $\mu\text{IU/mL}$), sensitivity analyses restricted to infants within the reference range (≤ 12 $\mu\text{IU/mL}$) yielded similar results (Table S10), suggesting that these cases did not materially influence the overall findings.

4.4. Limitations, Strengths, and Implications

Several limitations must be acknowledged. First, although placental type was included as a covariate, stratified analyses by chorionicity, i.e., dichorionic diamniotic (DCDA) vs monochorionic diamniotic (MCDA), were not feasible because MCDA pregnancies comprised only 6.4% of the population. This constrained our statistical power to fully capture potential differences in PFAS pharmacokinetics or placental pathology specific to monochorionic twins. Second, the relatively small sample size limited the ability to assess trimester-specific exposure patterns or to robustly link exposure timing to health outcomes, both of which are important for identifying critical windows of susceptibility during gestation. Third, the placental gene expression analyses were conducted in a relatively small subset of the population and should be considered to be exploratory. Moreover, given the inherent multicollinearity between PFAS exposure and MVU severity, these analyses remain preliminary, warranting further validation in larger cohorts to elucidate their independent effects. Fourth, although relatively uniform delivery conditions and the twin design likely limited variability and its potential influence, the lack of information on the timing of cord clamping remains a limitation, as it may influence cord-blood thyroid hormone concentrations.⁵⁵ Fifth, the study primarily examined individual PFAS compounds without evaluating potential combined or synergistic mixture effects, which more closely reflect real-world exposures. Sixth, maternal medication use during pregnancy was not fully accounted for in the analyses; therefore, residual confounding cannot be entirely excluded. Certain medications may affect maternal thyroid hormones or metabolic processes involved in PFAS kinetics. Lastly, reverse causation cannot be entirely excluded, as impaired placental function may influence fetal PFAS concentrations, potentially acting as a mediator or collider in the pathway linking maternal PFAS exposure to thyroid hormone disturbances or fetal growth. Because cord PFAS levels largely reflect the maternal body burden, this possibility may be limited. These limitations warrant further investigation in larger cohorts with more detailed clinical data to better elucidate the complex relationships between PFAS exposure and maternal–fetal health in twin pregnancies.

Despite these limitations, this study has its own unique strengths. This is the first study in Korea to investigate PFAS exposure and associated health outcomes in twin pregnancies. The focus on twin pregnancies is a key advantage, as most prior PFAS research has been conducted in singleton pregnancies, limiting the understanding of exposure dynamics and health impacts in multiple gestations, which are becoming increasingly prevalent with advanced maternal age. The measurement of PFAS in both maternal and cord serum samples enabled a comprehensive assessment of prenatal exposure and its relationship to thyroid hormone regulation and placental pathologies. Furthermore, within-pair analyses provided unique insights into variability in intrauterine exposure, which can be examined only in twin studies.

This study shows that several PFAS, including PFOA, PFNA, and ΣPFCA , may compromise placental endocrine and vascular function in twin pregnancies, potentially reducing thyroid hormone availability and contributing to fetal growth restriction. Given the persistence of PFAS and the relatively high exposure levels observed in this population, these findings highlight the need for continued biomonitoring, further mechanistic investigation, and public health measures to mitigate prenatal PFAS exposure and protect maternal–fetal health.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.5c16517>.

Abbreviations and nomenclature of PFAS compounds and internal standards, detailed analytical conditions and mass spectrometry parameters using UPLC–MS/MS, accuracy and precision results for method validation, primer sequences for placental gene expression analysis, cohort characteristics, sensitivity and GEE analyses of associations between PFAS exposure, thyroid hormones, and fetal outcomes, odds ratios for maternal vascular underperfusion (MVU), associations with placental pathology and gene expression levels, and supporting figures for PFAS-associated thyroid, fetal, placental, and correlation analyses (PDF)

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Notes

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